

views::cache_last

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§ 1 Abstract

This paper proposes the `views::cache_last` adaptor that caches the result of the last dereference of the underlying iterator. To support this adaptor, it also proposes an exception and a clarification to 16.4.6.10 [\[res.on.data.races\]](#).

§ 2 Revision history

— R1: Limited the 16.4.6.10 [\[res.on.data.races\]](#) carve-out to this adaptor per SG1 feedback.

§ 3 Motivation

The motivation for this view is given in [\[P2760R1\]](#) and quoted below for convenience:

One of the adaptors that we considered for C++23 but ended up not pursuing was what range-v3 calls `cache1` and what we'd instead like to call something like `cache_last`. This is an adaptor which, as the name suggests, caches the last element. The reason for this is efficiency - specifically avoiding extra work that has to be done by iterator dereferencing.

The canonical example of this is `transform(f) | filter(g)`, where if you then iterate over the subsequent range, `f` will be invoked twice for every element that satisfies `g`:

```
int main()
{
    std::vector<int> v = {1, 2, 3, 4, 5};

    auto even_squares = v
        | std::views::transform([](int i){
            std::print("transform: {}\n", i);
            return i * i;
        })
        | std::views::filter([](int i){
            std::print("filter: {}\n", i);
            return i % 2 == 0;
        });
```

```

    for (int i : even_squares) {
        std::print("Got: {}\n", i);
    }
}

```

prints the following (note that there are 7 invocations of transform):

```

transform: 1
filter: 1
transform: 2
filter: 4
transform: 2
Got: 4
transform: 3
filter: 9
transform: 4
filter: 16
transform: 4
Got: 16
transform: 5
filter: 25

```

The solution here is to add a layer of caching:

```

auto even_squares = v
    | views::transform(square)
    | views::cache_last
    | views::filter(is_even);

```

Which will ensure that square will only be called once per element.

§ 4 Design

§ 4.1 Caching in `operator*`

This is the only reasonable place for this caching to happen. Caching in `operator++` would require unnecessary overhead on every traversal if the user does not need to dereference every iterator (e.g., a striding view).

§ 4.2 Relaxing `[res.on.data.races]`

Because `operator*` is required to be a `const` member function (the cost of relaxing that requirement would be prohibitive as it affects every iterator and range adaptor), yet we want to have it perform a potentially-modifying operation, our options are basically:

- Relax 16.4.6.10 [\[res.on.data.races\]](#) p3
- Require synchronization.

This paper proposes the former. Input-only iterators, in general, are poor candidates for sharing across threads because, even when copyable (and they are not required to be in C++20), they are subject to “spooky action at a distance” where incrementing one iterator invalidates all copies thereof. This is why parallel algorithms require at least forward iterators. Sharing references to the same input iterator across threads seems like a fairly contrived scenario.

Moreover, `std::istreambuf_iterator` already doesn’t meet this requirement in [every major standard library implementation](#) yet it does not seem to have appeared on any standard library maintainer’s radar in the decade since the publication of C++11. This further suggests that this guarantee is not relied upon in practice for input-only iterators.

It is true that we could require synchronization on `operator*() const`. This probably isn’t terrible expensive in this context (we only need to protect against `operator*` calls racing with each other, not them racing with `operator++`), but adding synchronization to an adaptor whose primary purpose is to improve performance seems particularly dissatisfying when that synchronization will almost never be actually necessary.

During the Tokyo SG1 meeting, the room favored a limited carve-out to 16.4.6.10 [\[res.on.data.races\]](#) for this adaptor only. As it turns out, p1 of that subclause already has “unless otherwise specified”, so we don’t need to make any additional modification there. However, the wording is unclear how it applies to templated functions in the standard library, so this paper proposes a clarification.

§ 4.3 What’s the reference type?

`range-v3` uses `range_value_t<V>&&`, but this somewhat defeats the purpose of caching if you can so easily invalidate it. Moreover, there doesn’t seem to be a reason to force an independent copy of the `value_type`. So long as the underlying iterator is not modified, the reference obtained from `operator*` should remain valid.

This paper therefore proposes `range_reference_t<V>&`. Note that even if the reference type is a real reference, it can be an *expensive-to-compute* reference, so caching could still make sense.

§ 4.4 Properties

`cache_last` is never borrowed, input-only, never common, and not const-iterable.

§ 4.5 `iter_move` and `iter_swap`

`indirectly_readable` and `indirectly_swappable` requires `iter_move` and `iter_swap` to respectively not modify `i` (in the 18.2 [\[concepts.equality\]](#) sense). Moreover, `indirectly_readable` requires `*i` to be equality-preserving. So the cache should not be invalidated by either operation. (The underlying element might be modified, but the reference itself, obtained from dereferencing the iterator, cannot.)

§ 5 Wording

This wording is relative to [\[N4971\]](#).

§ 5.1 Clarifying [res.on.data.races]

Edit 16.4.6.10 [res.on.data.races] p1 as indicated:

- 1 This subclause specifies requirements that implementations shall meet to prevent data races. Every standard library function shall meet each requirement unless otherwise specified. Implementations may prevent data races in cases other than those specified below. **For the purpose of applying these requirements to standard library templated functions, each operation on types dependent on a template argument is assumed to meet these requirements. [Note 1: If a supplied operation does not meet these requirements, a data race can result, but the behavior is otherwise well-defined. — end note]**
- 2 A C++ standard library function shall not directly or indirectly access objects ([intro.multithread]) accessible by threads other than the current thread unless the objects are accessed directly or indirectly via the function's arguments, including `this`.
- 3 A C++ standard library function shall not directly or indirectly modify objects (6.9.2 [intro.multithread]) accessible by threads other than the current thread unless the objects are accessed directly or indirectly via the function's non-const arguments, including `this`.

§ 5.2 Addition to <ranges>

Add the following to 26.2 [ranges.syn], header <ranges> synopsis:

```
// [...]
namespace std::ranges {
    // [...]

    // [range.cache.last], to input view
    template<input_range V>
        requires view<V>
        class cache_last_view;

    namespace views {
        inline constexpr unspecified cache_last = unspecified;
    }
}
```

§ 5.3 cache_last

Add the following subclause to 26.7 [range.adaptors]:

§ 26.7.? Cache last view [range.cache.last]

§ 26.7.?.1 Overview [range.cache.last.overview]

- 1 `cache_last_view` caches the last element of its underlying sequence so that the element does not have to be recomputed on repeated access. [Note 1: This is useful if computation of the element to produce is expensive. — end note]

- 2 The name `views::cache_last` denotes a range adaptor object (26.7.2 [\[range.adaptor.object\]](#)). Let `E` be an expression. The expression `views::cache_last(E)` is expression-equivalent to `cache_last_view(E)`. [Drafting note: Intentional CTAD to avoid double wrapping if `E` is already a `cache_last_view`.]

§ 26.7.2.2 Class template `cache_last_view` `[range.cache.last.view]`

```
template<input_range V>
    requires view<V>
class cache_last_view : public view_interface<cache_last_view<V>>{
    V base_ = V(); // exposition
    only
    using cache_t = conditional_t<is_reference_v<range_reference_t<V>>, // exposition
    only
                                add_pointer_t<range_reference_t<V>>,
                                range_reference_t<V>>;

    non-propagating-cache<cache_t> cache_; // exposition
    only

    class iterator; // exposition
    only
    class sentinel; // exposition
    only

public:
    cache_last_view() requires default_initializable<V> = default;
    constexpr explicit cache_last_view(V base);

    constexpr V base() const & requires copy_constructible<V> { return base_; }
    constexpr V base() && { return std::move(base_); }

    constexpr auto begin();
    constexpr auto end();

    constexpr auto size() requires sized_range<V>;
    constexpr auto size() const requires sized_range<const V>;
};

template<class R>
    cache_last_view(R&&) -> cache_last_view<views::all_t<R>>;
```

```
constexpr explicit cache_last_view(V base);
```

- 1 *Effects:* Initializes `base_` with `std::move(base)`.

```
constexpr auto begin();
```

- 2 *Effects:* Equivalent to:

```
    return iterator(*this);
```

```
constexpr auto end();
```

- 3 *Effects:* Equivalent to:

```
return sentinel(*this);
```

```
constexpr auto size() requires sized_range<V>;  
constexpr auto size() const requires sized_range<const V>;
```

4 *Effects*: Equivalent to:

```
return ranges::size(base_);
```

§ 26.7.?.3 Class `cache_last_view::iterator` [range.cache.last.iterator]

```
namespace std::ranges {  
    template<input_range V>  
        requires view<V>  
    class cache_last_view<V>::iterator {  
        cache_last_view* parent_;           // exposition only  
        iterator_t<V> current_;             // exposition only  
  
        constexpr explicit iterator(cache_last_view& parent); // exposition only  
  
    public:  
        using difference_type = range_difference_t<V>;  
        using value_type = range_value_t<V>;  
        using iterator_concept = input_iterator_tag;  
  
        iterator(iterator&&) = default;  
        iterator& operator=(iterator&&) = default;  
  
        constexpr iterator_t<V> base() &&;  
        constexpr const iterator_t<V> base() const & noexcept;  
  
        constexpr range_reference_t<V>& operator*() const;  
  
        constexpr iterator& operator++();  
        constexpr void operator++(int);  
  
        friend constexpr range_rvalue_reference_t<V> iter_move(const iterator& i)  
            noexcept(noexcept(ranges::iter_move(i.current_)));  
  
        friend constexpr void iter_swap(const iterator& x, const iterator& y)  
            noexcept(noexcept(ranges::iter_swap(x.current_, y.current_)))  
            requires indirectly_swappable<iterator_t<V>>;  
    };  
}
```

```
constexpr explicit iterator(cache_last_view& parent);
```

1 *Effects*: Initializes `current_` with `ranges::begin(parent.base_)` and `parent_` with `addressof(parent)`.

```
constexpr iterator_t<V> base() &&;
```

2 *Returns*: `std::move(current_)`.

```
constexpr const iterator_t<V>& base() const & noexcept;
```

3 *Returns:* `current_`.

```
constexpr iterator& operator++();
```

4 *Effects:* Equivalent to:

```
++current_;  
parent_>cache_.reset();  
return *this;
```

```
constexpr void operator++(int);
```

5 *Effects:* Equivalent to: `++*this`;

```
constexpr range_reference_t<V>& operator*() const;
```

6 *Effects:* Equivalent to:

```
if constexpr (is_reference_v<range_reference_t<V>>) {  
    if (!parent_>cache_) {  
        parent_>cache_ = addressof(as_lvalue(*current_));  
    }  
    return **parent_>cache_;  
}  
else {  
    if (!parent_>cache_) {  
        parent_>cache_.emplace_deref(current_);  
    }  
    return *parent_>cache_;  
}
```

7 [*Note 1:* Evaluations of `operator*` on the same iterator object may conflict (6.9.2.2 [\[intro.races\]](#)). — *end note*]

```
friend constexpr range_rvalue_reference_t<V> iter_move(const iterator& i)  
    noexcept(noexcept(ranges::iter_move(i.current_)));
```

8 *Effects:* Equivalent to: `return ranges::iter_move(i.current_);`

```
friend constexpr void iter_swap(const iterator& x, const iterator& y)  
    noexcept(noexcept(ranges::iter_swap(x.current_, y.current_)))  
    requires indirectly_swappable<iterator_t<V>>;
```

9 *Effects:* Equivalent to: `ranges::iter_swap(x.current_, y.current_);`

§ 26.7.?.3 Class `cache_last_view::sentinel` [`range.cache.last.sentinel`]

```
namespace std::ranges {  
    template<input_range V>  
        requires view<V>  
        class cache_last_view<V>::sentinel {  
            sentinel_t<V> end_ = sentinel_t<V>(); // exposition only  
  
            constexpr explicit sentinel(cache_last_view& parent); // exposition only  
        }  
};
```

```

public:
    sentinel() = default;

    constexpr sentinel_t<V> base() const;

    friend constexpr bool operator==(const iterator& x, const sentinel& y);
};

```

```
constexpr explicit sentinel(cache_last_view& parent);
```

¹ *Effects:* Initializes `end_` with `ranges::end(parent.base_)`.

```
constexpr sentinel_t<V> base() const;
```

² *Returns:* `end_`.

```
friend constexpr bool operator==(const iterator& x, const sentinel& y);
```

³ *Returns:* `x.current_ == y.end_;`

§ 6 References

[N4971] Thomas Köppe. 2023-12-18. Working Draft, Programming Languages — C++.

<https://wg21.link/n4971>

[P2760R1] Barry Revzin. 2023-12-15. A Plan for C++26 Ranges.

<https://wg21.link/p2760r1>